Chemistry · Unit 1B · Level 2 (modified)

Electric Forces



TODAY'S PLAN

- 🧲 Attract and repel
- 📐 Coulomb's law
- 🔢 Big numbers
- 🔀 Neutral objects
- 📝 Review

Review · 5 minutes

Quick Review



➖ Rubbing moves **electrons**.

🎈 Extra electrons = **negative**; missing electrons = **positive**.

❓ What happens when two rubbed balloons get close?

Words to Know

Words to Know



Attract

Pull toward each other. Opposite charges (+ and −) attract.



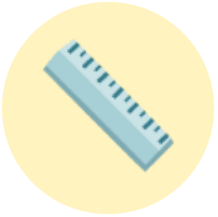
Repel

Push away from each other. Like charges (− and −, or + and +) repel.



Electric force

The push or pull between charged objects. It works without touching.



Distance

How far apart two objects are. Closer = stronger force.

Words to Know

More Words to Know



🔋 **Amount of charge**: How many extra or missing electrons. More charge = stronger force.

🔀 **Polarized**: Electrons in a neutral object shift to one side, so one side is a little - and the other a little +.

📐 **Coulomb's law**: The rule for electric force: more charge makes it stronger; more distance makes it much weaker.

🔢 **Scientific notation**: A short way to write very big or small numbers, like $2 \times 10^{12} = 2{,}000{,}000{,}000{,}000$.

Compare

Attract or Repel



🧲

Attract

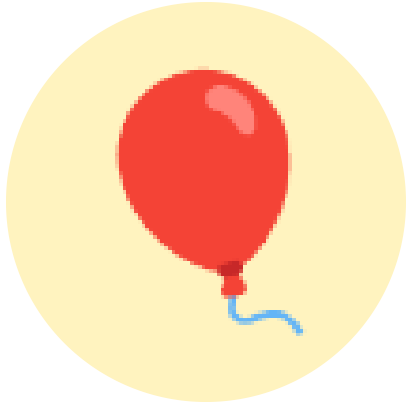
Opposite charges ➕ ➖ pull together

↔️

Repel

Like charges ➖ ➖ or ➕ ➕ push apart

🃏 Check 1 of 4



Two balloons, both rubbed on hair, will…

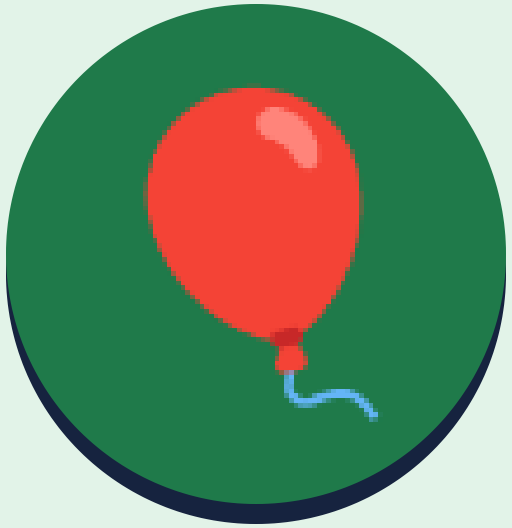
Hold up 1, 2, or 3:

1️⃣ Attract

2️⃣ Repel

3️⃣ Do nothing

✅ Answer · Card 1 of 4



Repel

Both are negative. Like charges repel.

Learn

Coulomb's Law: Two Rules

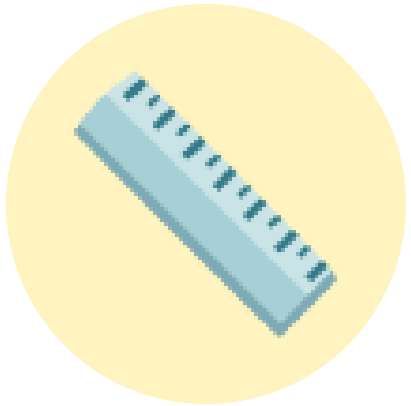


🔋 More **charge** → **stronger** force.

📏 More **distance** → **weaker** force.

⚠️ Distance matters a lot: **2 times** farther → only **¼** the force.

🃏 Check 2 of 4



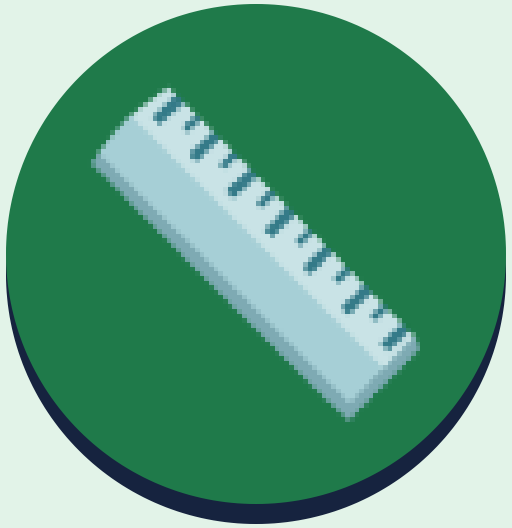
You move two charged objects farther apart. The force…

Hold up 1, 2, or 3:

1️⃣ Gets stronger

2️⃣ Stays the same

3️⃣ Gets weaker



Gets weaker

More distance means weaker force.

🃏 Check 3 of 4



You add more charge to one balloon. The force...

Hold up 1, 2, or 3:

1️⃣ Gets stronger

2️⃣ Gets weaker

3️⃣ Disappears

✅ Answer · Card 3 of 4



Gets stronger

More charge means stronger force.

Learn

Big Numbers: Scientific Notation



🔢 A balloon can hold about **2,000,000,000,000** extra electrons.

✏️ Short way: 2×10^{12} (the 12 means 12 zeros).

📈 A bigger power of 10 means a **bigger** number.

Learn

Neutral Objects Can Be Pulled

📄 Paper scraps are **neutral**.

🎈 A negative balloon pushes the paper's **electrons** to the far side.

🔀 The near side is now a little **+**, so the paper is **pulled** to the balloon (polarized).

🙌 No touching needed!



🃏 Check 4 of 4



A charged balloon pulls on neutral paper because the paper's electrons…

Hold up 1, 2, or 3:

1️⃣ Leave the paper

2️⃣ Turn into protons

3️⃣ Shift to the far side

✅ Answer · Card 4 of 4



Shift to the far side

The near side becomes slightly +.

Wrap-Up

Recap



Opposites
attract



Like charges
repel



Coulomb's law
more charge, less distance = stronger



Polarized
neutral object gets pulled

Wrap-Up

Exit Ticket



1️⃣ What two things make the electric force stronger?

2️⃣ Write 3×10^4 as a regular number.

3️⃣ Why does a balloon pick up neutral paper?